THE KERNEL OF A HOMOLOGICAL FUNCTOR AND ITS MULTIPLICATIVE SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2465-2489, following the statement at 2447-2463.

Editorial note. No omissions within this span: the LES identification is the only mathematical step and it is read off the displayed long exact sequence.

Allowed standard prerequisites (editorial).

- Kernel of a homological functor. lemma-homological-functor-kernel (same chapter, cited at ~2468; a published-range-gated span used here as citation only). Ker(H) is a strictly full saturated triangulated subcategory.
- Multiplicative system of a homological functor. lemma-homological-functor-localize (same chapter, cited at ~2471). S is a saturated compatible multiplicative system.
- Universal property of the quotient. lemma-universal-property-quotient (same chapter, cited at ~2488). Factors H through the quotient.

Source reference. Original definition 0147: sources/stacks-derived/pinned-original-source.tex, source lines 478–478. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0148: sources/stacks-derived/pinned-original-source.tex, source lines 499–499. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05RD: sources/stacks-derived/pinned-original-source.tex, source lines 2030–2030. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05R5: sources/stacks-derived/pinned-original-source.tex, source lines 1525–1525. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05RL: sources/stacks-derived/pinned-original-source.tex, source lines 2395–2395. Navigation pointer only: the counted proof

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cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05RJ: sources/stacks-derived/pinned-original-source.tex, source lines 2303–2303. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $H : \mathcal{D} \to \mathcal{A}$ be a homological functor from a triangulated category $\mathcal{D}$ to an abelian category $\mathcal{A}$, see Definition 0147. The subcategory $\mathrm{Ker}(H)$ of $\mathcal{D}$ is a strictly full saturated triangulated subcategory of $\mathcal{D}$ whose corresponding saturated multiplicative system (see Lemma 05RL) is the set*

$$S = \{f \in \mathit{Arrows}(\mathcal{D}) \mid H^i(f) \text{ is an isomorphism for all } i \in \mathbf{Z}\}.$$

The functor H factors through the quotient functor $Q : \mathcal{D} \to \mathcal{D}/\mathrm{Ker}(H)$.

Proof. The category $\mathrm{Ker}(H)$ is a strictly full saturated triangulated subcategory of $\mathcal{D}$ by Lemma 05RD. The set S is a saturated multiplicative system compatible with the triangulated structure by Lemma 05R5. Recall that the multiplicative system corresponding to $\mathrm{Ker}(H)$ is the set

$$\left\{ \begin{array}{c} f \in \mathrm{Arrows}(\mathcal{D}) \text{ such that there exists a distinguished triangle} \\ (X, Y, Z, f, g, h) \text{ with } H^i(Z) = 0 \text{ for all } i \end{array} \right\}.$$

By the long exact cohomology sequence, see (0148), it is clear that f is an element of this set if and only if f is an element of S. Finally, the factorization of H through Q is a consequence of Lemma 05RJ. □